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Assignment 1
Integrated Energy Grids

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1 Section A: Optimal Capacities for One Country

Germany was selected as the case study for the year 2024. The electricity system was modeled as a single-node PyPSA network with one electricity bus representing the national system. Four generation technologies were included: onshore wind, offshore wind, solar PV, and gas. Techno-economic assumptions were taken from [1] (Table 3). Historical generation and installed capacity data were obtained from Energy-Charts [2], [3]. Renewable capacity factors were calculated as:

$$CF_t^{\text{tech}} = \frac{\text{generation}_t^{\text{tech}}}{\text{installed capacity}_{\text{year}}^{\text{tech}}}, \tag{1}$$

Annualized capital costs were computed using:

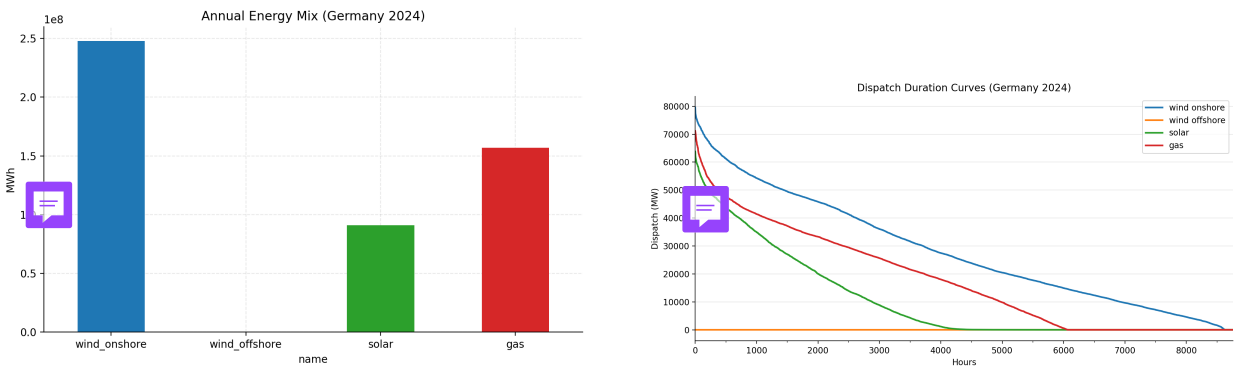
$$a(r,n) = \frac{r}{1-(1+r)^{-n}}, \qquad CapitalCost_{\text{tech}} = a(r,n)\, CAPEX_{\text{tech}}\,(1+FOM_{\text{tech}}), \tag{2}$$

We assumed renewable generators to have zero marginal cost, and for gas generation:

$$MC_{\text{gas}} = \frac{c_{\text{gas}}}{\eta_{\text{gas}}} + p_{\text{CO}_2}\frac{e_{\text{gas}}}{\eta_{\text{gas}}}, \tag{3}$$

A CO_2 price of 84.15 €/tCO_2 was used, obtained as the average of the reported range (90–103 USD/tCO_2) in [4] and converted to EUR.

The optimization determines the cost-optimal capacities required to meet annual demand. Results (Table 4) show a system dominated by onshore wind, with solar PV also contributing significantly, while offshore wind is not selected. Gas provides dispatchable backup capacity. Figures 1a–1b show the operational roles, onshore wind supplies a large share of generation across many hours, solar PV is concentrated in daytime (especially in summer), and gas flexibly covers residual demand, particularly during periods of low renewable availability. The absence of offshore wind is due to its higher capital cost relative to the other generation alternatives.



(a) Annual electricity mix

(b) Duration curves

Figure 1: Annual production and duration curves for the optimized German 2024 system

2 Section B: Interannual variability

To investigate the impact of interannual variability, hourly capacity factors for solar, onshore wind, and offshore wind were computed for each year from 2015–2025 using the installed capacities of each year (Appendix Table 5) and Equation (1). Electricity demand was fixed to the 2024 profile in order to analyse how the demand of that year could be met under the different weather condition. The evolution of annual capacity factors is shown in Figure 10: solar remains relatively stable over time, while offshore wind shows a higher capacity factors than onshore wind. The model setup and cost assumptions follow Section 1.

Results The main takeaways are:

- **Offshore wind** is not selected in any year due to its high capital cost relative to other alternatives.
- **Gas** shows little variability, indicating a stable role as a backup capacity.
- **Solar and onshore wind** show higher variability, reflecting high sensitivity to weather conditions; years with lower resource availability leads to higher required installed capacities, and vice versa. This shows the importance of accounting for interannual variability in capacity planning.

Figures 2, 3 and Table 1 summarise the optimal capacities and their variability across years.

Technology	Mean Capacity (MW)	Std Deviation (MW)
Wind Onshore	167348.90	6128.53
Wind Offshore	0.00	0.00
Solar	118314.85	5315.34
Gas	72652.68	1629.14

Table 1: Mean optimal capacities and variability between 2015 and 2025

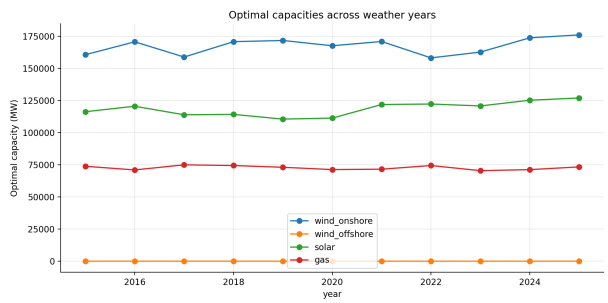


Figure 2: Optimal installed capacities (2015–2025)

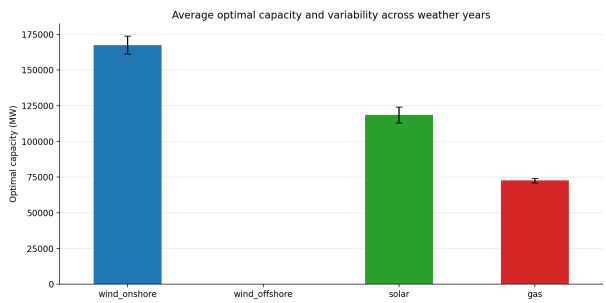


Figure 3: Average optimal capacity and variability across years

3 Section C: Storage Addition

A battery energy storage system (BESS) is added to the model to investigate how storage affects the optimal system configuration and helps balance renewable generation. The techno-economic parameters of the BESS are taken from [1] (see Table 6). We built from the same system configuration and cost assumptions as in Section 1. The total investment cost per MW of battery power capacity is calculated as:

$$C_{\text{BESS}} = C_{\text{energy}} \cdot h_{\max} + C_{\text{inverter}} \tag{4}$$

Where C_{energy} is the cost per MWh of battery energy capacity, and C_{inverter} is the cost per MW of inverter power capacity.

Results The resulting optimal battery has a power capacity of 850 MW and an energy capacity of 5110 MWh, a 6-hour system. The following patterns can be observed:

- **Seasonal:** The battery utilization is higher in summer than in winter, driven by increased solar PV generation (Figures 11, 5).
- **Intraday:** The battery charges during midday when solar generation is abundant and discharges during evening and night hours, indicating generation shifting. If prices were modelled, the battery would have been able to take advantage of the usual cheap electricity prices that happen midday to charge and evening higher prices to discharge. See Figure 4 to see the intraday patterns.
- **Battery usage:** The battery is actively cycled throghout the year, clearly confirming its role in balancing energy availability and variability (Figures 12, 13).

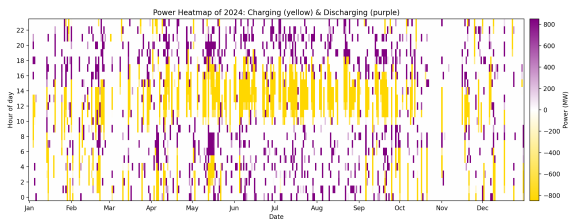


Figure 4: Hourly battery charging and discharging power over the year

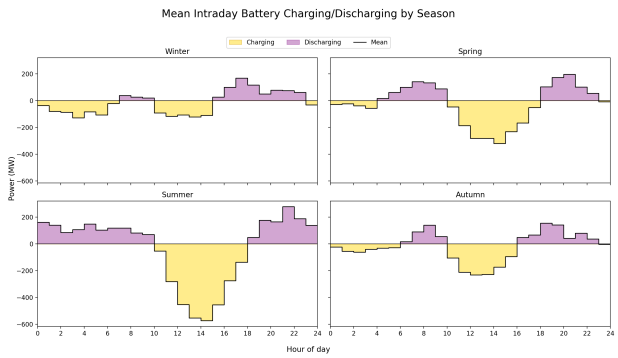


Figure 5: Mean intraday battery charging and discharging profile by season

4 Section D: Connection to Neighbouring Countries

To analyse the impact of cross-border transmission, the German system was extended to a four-node network including Belgium, France, and the Netherlands. Each country was represented by a single bus. The interconnections included were:

DE–BE, DE–FR, DE–NL, BE–FR, and BE–NL

ensuring at least the one closed cycle: DE $\rightarrow$ BE $\rightarrow$ FR $\rightarrow$ DE .

All lines were modelled as HVAC interconnectors with a nominal voltage of 400 kV and a fixed reactance of $x = 0.1$. The network was solved using the linearised AC power flow approximation in PyPSA.

Germany was modelled with the same technologies as in previous sections (onshore wind, offshore wind, solar PV, gas, and battery storage). For Belgium, France, and the Netherlands, installed capacities were fixed to the 2024 values. A residual generation technology was included to represent non-modelled generation (nuclear, hydro, etc...), defined as the difference between total generation and the explicitly modelled wind, solar, and gas. Transmission capacities were fixed using observed cross-border flows in 2024. For each interconnector, the maximum absolute hourly flow was used as a proxy for the line capacity. While this does not represent the exact technical limit, it provides a data-driven approximation consistent with historical operation.

Country-specific marginal costs were assigned to residual generation to reflect differences in generation mix. In France, the residual mainly represents nuclear and hydropower and was assigned a low cost of 12 EUR/MWh [5]. Belgium has a similar but more mixed structure, leading to an intermediate value of 22 EUR/MWh [6]. In the Netherlands, the residual includes a higher share of thermal generation, so a higher cost of 45 EUR/MWh was assumed [7].

Compared to the previous isolated system, interconnection provides additional flexibility to the system: Germany can import electricity during periods of low renewable availability and export during periods of high generation.

Results The optimal solution shows us that interconnection with other countries significantly affects the optimal German capacity mix. In particular:

- **Onshore wind** becomes the dominant technology with approximately 160.9 GW installed,
- **Solar PV** reaches around 105.9 GW of installed capacity,
- **Gas** capacity is reduced to about 56.4 GW, and
- **Offshore wind** is still not deployed.

This indicates that access to neighbouring systems reduces the need for dispatchable gas backup and shifts the system towards higher reliance on onshore wind and solar. In Table 2,

we show the most relevant operating hours for the German system, highlighting the periods with maximum imports, exports, solar dispatch, residual dispatch, and line loading. From it, we can extract that the maximum import and export hours both reach 10 395 MW, showing that cross-border exchange frequently uses the available line capacity. Additionally, the load of the line reaches 100% during several hours. The hours of maximum solar dispatch happen during summer as expected, showing as well, that high renewable availability incurs potential exports, whereas the maximum import hours are found during winter, when lower renewable capacity is available. Overall, interconnection reduces system costs and improves the integration of renewable energy.

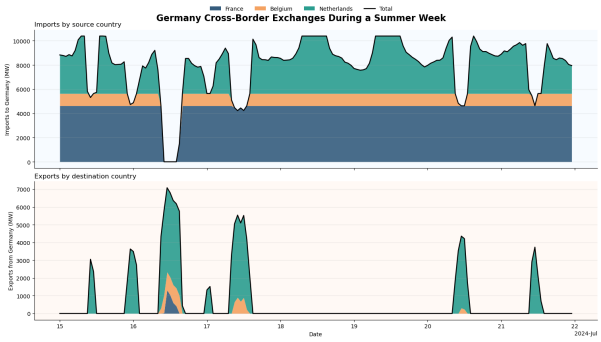


Figure 6: Cross-border exchange during summer week

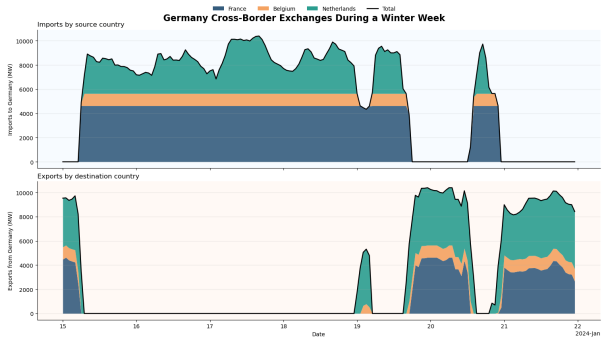


Figure 7: Cross-border exchange during winter week

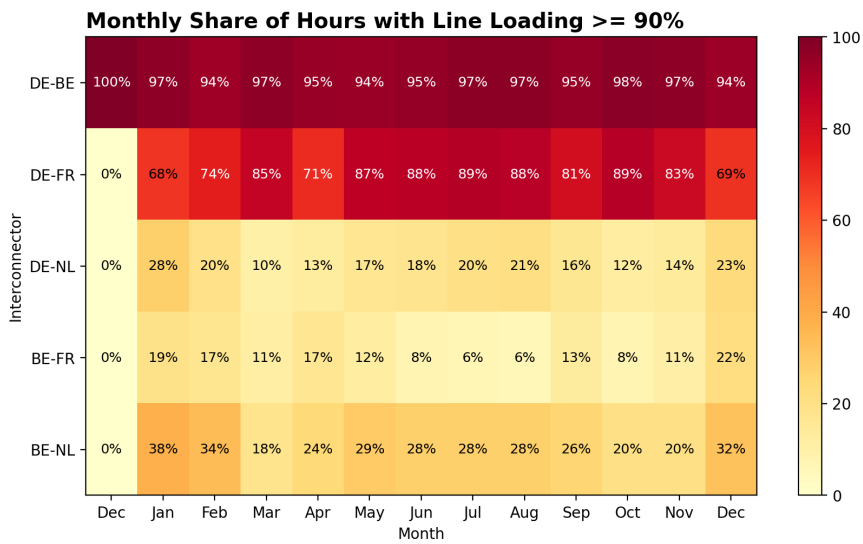


Figure 8: Monthly Share of hours with line loading >= 90percent

Top import hours	
Snapshot	**MW**
2024-01-17 16:00	10395.0
2024-01-27 16:00	10395.0
2024-01-27 17:00	10395.0
2024-02-11 15:00	10395.0
2024-02-11 16:00	10395.0
2024-02-11 17:00	10395.0
2024-02-11 18:00	10395.0
2024-02-22 07:00	10395.0

Top export hours	
Snapshot	**MW**
2024-01-02 17:00	10395.0
2024-01-03 16:00	10395.0
2024-01-03 17:00	10395.0
2024-01-13 22:00	10395.0
2024-01-19 23:00	10395.0
2024-01-20 06:00	10395.0
2024-01-20 07:00	10395.0
2024-01-22 07:00	10395.0

Top residual dispatch	
Snapshot	**MW**
2023-12-31 23:00	0.0
2024-01-01 00:00	0.0
2024-01-01 01:00	0.0
2024-01-01 02:00	0.0
2024-01-01 03:00	0.0
2024-01-01 04:00	0.0
2024-01-01 05:00	0.0
2024-01-01 06:00	0.0

Top solar dispatch	
Snapshot	**MW**
2024-06-25 11:00	46137.0
2024-07-29 11:00	46005.0
2024-06-25 10:00	45786.1
2024-07-30 11:00	45325.1
2024-07-29 10:00	45313.5
2024-07-30 10:00	45238.1
2024-07-09 11:00	45014.7
2024-07-29 12:00	44825.5

Top line loading	
Snapshot	**%**
2023-12-31 23:00	100.0
2024-01-01 00:00	100.0
2024-01-01 01:00	100.0
2024-01-01 02:00	100.0
2024-01-01 03:00	100.0
2024-01-01 04:00	100.0
2024-01-01 05:00	100.0
2024-01-01 06:00	100.0

Table 2: Key operating hours for Germany under interconnected system

5 Section E: Manual Calculation of Line Flows at $t = 1$

The aim of this section is to calculate the line flows within the network manually for the first time step ($t = 1$). The network buses are ordered as: $[DE,\ BE,\ FR,\ NL]$ and the lines, with their positive flow directions:

$$DE \to BE, \qquad DE \to FR, \qquad DE \to NL, \qquad BE \to FR, \qquad BE \to NL$$

All lines have reactance $x = 0.1$ and therefore susceptance $\frac{1}{x} = \frac{1}{0.1} = 10$.
The nodal imbalance vector at the first time step is:

$$\mathbf{p} = \begin{bmatrix} I_{DE} \\ I_{BE} \\ I_{FR} \\ I_{NL} \end{bmatrix} = \begin{bmatrix} 6514.40 \\ 2466.40 \\ -3113.18 \\ -5867.63 \end{bmatrix} \text{ MW} \tag{5}$$

which must satisfy:

$$I_{DE} + I_{BE} + I_{FR} + I_{NL} = 0 \tag{6}$$

Matrices

The incidence matrix is formed with $+1$ at the sending bus and -1 at the receiving bus:

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 & 1 \\ 0 & -1 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & -1 \end{bmatrix} \tag{7}$$

and the line susceptance matrix is:

$$B_l = \begin{bmatrix} 10 & 0 & 0 & 0 & 0 \\ 0 & 10 & 0 & 0 & 0 \\ 0 & 0 & 10 & 0 & 0 \\ 0 & 0 & 0 & 10 & 0 \\ 0 & 0 & 0 & 0 & 10 \end{bmatrix} \tag{8}$$

The bus susceptance matrix, also known as the weighted Laplacian matrix is then:

$$B_{\text{bus}} = AB_lA^\top = \begin{bmatrix} 30 & -10 & -10 & -10 \\ -10 & 30 & -10 & -10 \\ -10 & -10 & 20 & 0 \\ -10 & -10 & 0 & 20 \end{bmatrix} \tag{9}$$

Method 1: Voltage-angle method Because there are only three independent variables in Equation 6, DE is chosen as the slack bus:

$$V_{DE} = 0 \Rightarrow \begin{bmatrix} I_{BE} \\ I_{FR} \\ I_{NL} \end{bmatrix} = \begin{bmatrix} 30 & -10 & -10 \\ -10 & 20 & 0 \\ -10 & 0 & 20 \end{bmatrix} \begin{bmatrix} V_{BE} \\ V_{FR} \\ V_{NL} \end{bmatrix} \tag{10}$$

By substituting the injection values, we have:

$$\begin{bmatrix} 2466.4 \\ -3113.18 \\ -5867.63 \end{bmatrix} = \begin{bmatrix} 30 & -10 & -10 \\ -10 & 20 & 0 \\ -10 & 0 & 20 \end{bmatrix} \begin{bmatrix} V_{BE} \\ V_{FR} \\ V_{NL} \end{bmatrix} \tag{11}$$

$$V_{DE} = 0, \qquad V_{BE} = -101.20, \qquad V_{FR} = -206.26, \qquad V_{NL} = -343.98 \tag{12}$$

The line flows are obtained from:

$$f_{ij} = \frac{1}{x_{ij}}(V_i - V_j) \tag{13}$$

$$I_{DE\to BE} = 10(V_{DE} - V_{BE}) = 10(0 - (-101.20)) = 1012.00 \text{ MW} \tag{14}$$
$$I_{DE\to FR} = 10(V_{DE} - V_{FR}) = 10(0 - (-206.26)) = 2062.59 \text{ MW} \tag{15}$$
$$I_{DE\to NL} = 10(V_{DE} - V_{NL}) = 10(0 - (-343.98)) = 3439.8 \text{ MW} \tag{16}$$
$$I_{BE\to FR} = 10(V_{BE} - V_{FR}) = 10(-101.2 - (-206.26)) = 1050.59 \text{ MW} \tag{17}$$
$$I_{BE\to NL} = 10(V_{BE} - V_{NL}) = 10(-101.2 - (-343.98)) = 2427.81 \text{ MW} \tag{18}$$

Thus yielding:

$$\mathbf{f} = \begin{bmatrix} 1012.00 \\ 2062.59 \\ 3439.81 \\ 1050.59 \\ 2427.81 \end{bmatrix} \text{ MW} \tag{19}$$

Method 2: PTDF matrix method We compute the PTDF matrix, taking DE as the slack bus again, and its designated column in the matrix is set to 0.

$$\text{PTDF} = B_l A_{\text{red}}^{\top} B_{\text{bus,red}}^{-1} = \begin{bmatrix} 0 & -\frac{1}{2} & -\frac{1}{4} & -\frac{1}{4} \\ 0 & -\frac{1}{4} & -\frac{5}{8} & -\frac{1}{8} \\ 0 & -\frac{1}{4} & -\frac{1}{8} & -\frac{5}{8} \\ 0 & \frac{1}{4} & -\frac{3}{8} & \frac{1}{8} \\ 0 & \frac{1}{4} & \frac{1}{8} & -\frac{3}{8} \end{bmatrix}. \tag{20}$$

Therefore:

$$\mathbf{f} = \text{PTDF} \cdot \mathbf{p} = \begin{bmatrix} 0 & -\frac{1}{2} & -\frac{1}{4} & -\frac{1}{4} \\ 0 & -\frac{1}{4} & -\frac{5}{8} & -\frac{1}{8} \\ 0 & -\frac{1}{4} & -\frac{1}{8} & -\frac{5}{8} \\ 0 & \frac{1}{4} & -\frac{3}{8} & \frac{1}{8} \\ 0 & \frac{1}{4} & \frac{1}{8} & -\frac{3}{8} \end{bmatrix} \cdot \begin{bmatrix} 6514.40 \\ 2466.40 \\ -3113.18 \\ -5867.63 \end{bmatrix} = \begin{bmatrix} 1012.00 \\ 2062.59 \\ 3439.81 \\ 1050.59 \\ 2427.81 \end{bmatrix} \text{ MW} \tag{21}$$

These manually calculated results match the results obtained from PyPSA, which are summarised in Table 7.

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References

[1] M. Victoria, K. Zhu, T. Brown, G. B. Andresen, and M. Greiner, "The role of storage technologies throughout the decarbonisation of the sector-coupled European energy system," *Energy Conversion and Management*, vol. 201, p. 111 977, Dec. 2019, ISSN: 0196-8904. DOI: 10.1016/j.enconman.2019.111977. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0196890419309835.

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6 Appendix

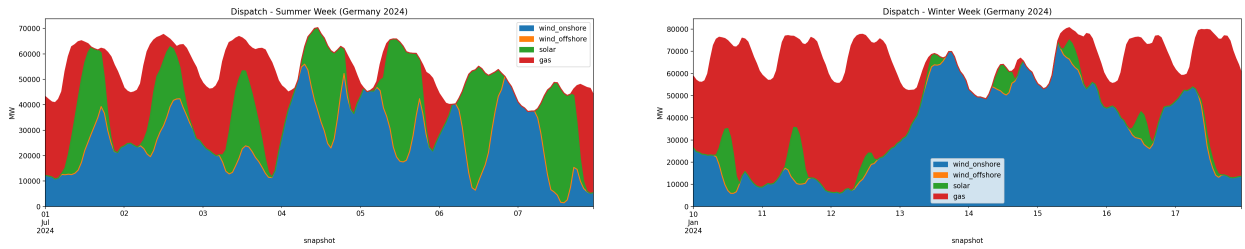
6.1 Section A: Figures and Tables

Technology	CAPEX [EUR/kW]	Lifetime [years]	Fixed O&M [%/y]	Efficiency
Onshore wind	910	30	0.033	–
Offshore wind	2506	25	0.030	–
Solar PV	425	25	0.030	–
Gas (OCGT)	560	25	0.033	0.39

Table 3: Main techno-economic assumptions used in the model

Technology	Optimal Capacity [MW]	Annual Generation [MWh]	Capacity Factor
Onshore wind	173875.57	2.476×10^8	0.162
Offshore wind	0.00	0.00	–
Solar PV	125259.03	9.089×10^7	0.082
Gas	71191.36	1.569×10^8	0.251

Table 4: Optimal capacities, annual generation, and Capacity Factor for German 2024 system



(a) Summer week dispatch

(b) Winter week dispatch

Figure 9: Dispatch of representative summer and winter weeks in Germany 2024

6.2 Section B: Figures and Tables

Year	Solar	Onshore Wind	Offshore Wind
2015	36970	41300	3380
2016	38330	45280	4240
2017	39860	50170	5520
2018	42580	52330	6510
2019	46230	53190	7660
2020	50920	54410	7870
2021	56070	56050	7870
2022	62920	58010	8220
2023	76920	61010	8470
2024	100688	63589	9215
2025	107400	68220	9730

Table 5: Installed renewable capacities (MW) in Germany

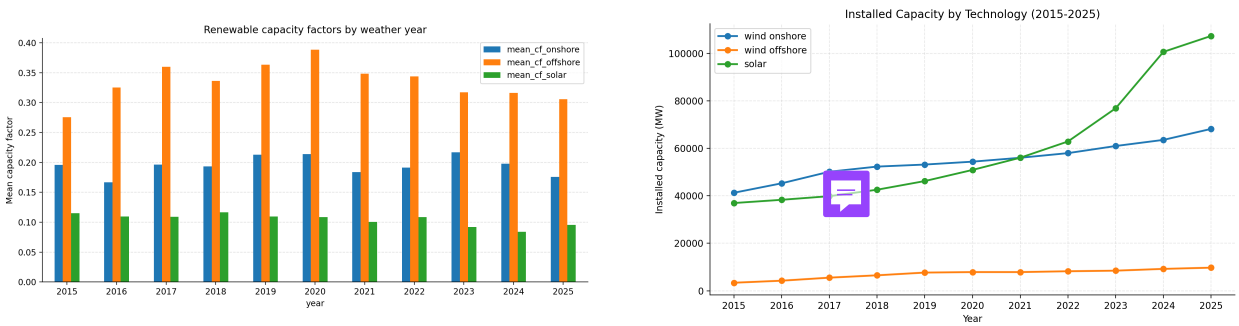


Figure 10: Renewable capacity factors and installed capacities from 2015 to 2025

6.3 Section C: Figures and Tables

Symbol	Parameter	Value
n	Lifetime	15 years
$h_{\max}$	Maximum storage duration	6 h
η_{ch}	Charging efficiency	0.90
η_{dis}	Discharging efficiency	0.90
η_{rt}	Round-trip efficiency	0.81
C_{energy}	Energy CAPEX	144 600 EUR/MWh
C_{inverter}	Inverter CAPEX	310 000 EUR/MW
FOM	Fixed O&M	3% of CAPEX
r	Discount rate	7%

Table 6: BESS parameters used in the model

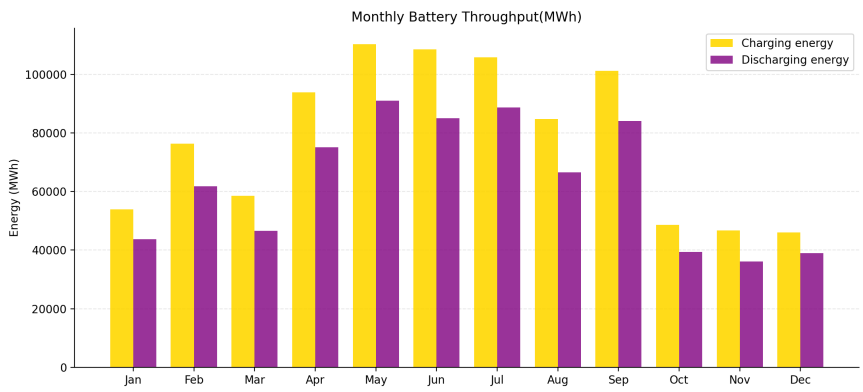


Figure 11: Monthly battery charging and discharging throughput

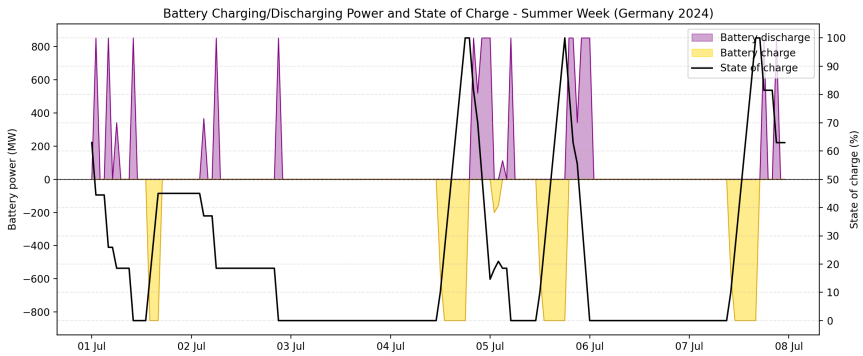


Figure 12: Battery charging, discharging, and state of charge during a summer week

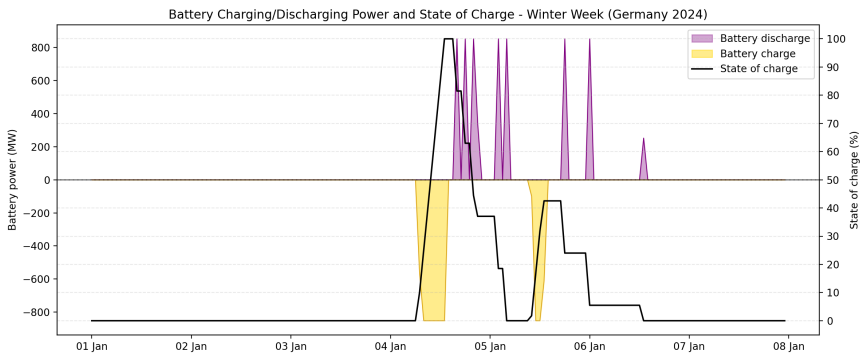


Figure 13: Battery charging, discharging, and state of charge during a winter week.

6.4 Section E: Figures and Tables

Bus	Generation [MW]	Load [MW]	Imbalance [MW]
Germany	45333.625	38818.125	6515.500
Belgium	9802.800	7335.300	2467.500
France	48501.050	51600.000	-3098.950
Netherlands	5358.100	11242.150	-5884.050
Line			**Flow [MW]**
Germany–Belgium			1012.000
Germany–France			2055.475
Germany–Netherlands			3448.025
Belgium–France			1043.475
Belgium–Netherlands			2436.025

Table 7: PyPSA results for $t = 1$, 2024-01-01 00:00:00